

How acne medications work

- Seeing change in acne usually takes 3–4 weeks and may take longer with severe or nodular acne.
- **Benzoyl peroxide** is highly effective at killing bacteria that contribute to acne and helps prevent new pimples from forming.
- **Topical retinoids** (adapalene, tazarotene, tretinoin) unclog pores, preventing acne development beneath the skin's surface. They also reduce redness, swelling, and inflammation associated with pimples.
- **Antibiotics** (clindamycin, erythromycin) work by killing acne-causing bacteria.
- **Hormonal therapies** for women reduce oil production triggered by androgens.
- **Oral isotretinoin** works through multiple mechanisms: it unclogs pores, reduces sebum (oil) production, inhibits bacterial growth, and decreases inflammation.

Best way to apply topical medications

- Use a pea-sized amount of medication and spread a thin layer over each affected area of the face.
- Using more will not speed up results and may increase the risk of irritation.
- Stick to the recommended amount and maintain consistent application.
- Do not apply only to individual spots ("spot treat")—apply to all acne-prone areas as directed.

Potential side-effects of topical acne treatments

- Common side effects include redness, scaling, dryness, stinging, and/or burning.
- These are typically mild and tend to improve after the first few weeks of treatment.
- Using a non-comedogenic moisturizer can help manage dryness.

Potential side-effects of oral acne treatments

Acne skin care Dos

- Wash your face once daily with a mild, soap-free, fragrance-free cleanser.
- For oily skin, choose a more detergent-like cleanser to reduce surface oil.
- For dry or sensitive skin, use a moisturizing cleanser.
- Apply moisturizer 2–3 times per day if you have dry skin.
- Use a richer cream or lotion in fall and winter; switch to a lighter formula in spring and summer.
- Use daily sunscreen with SPF 25–50 to prevent dark spots, discoloration, and premature aging.
- Consult a dermatologist if you develop irritation or other skin concerns.

Acne skin care Don'ts

- Avoid aggressive or frequent cleansing—scrubbing is unnecessary and can worsen acne.

- Avoid alcohol-based cleansers, which can irritate the skin and aggravate acne.
- Do not use scented products if you have sensitive skin.
- Herbal or floral ingredients are not necessarily better and may cause irritation.
- Avoid topical vitamin A and retinol products, as they can be drying; also avoid greasy products like petroleum jelly due to their pore-clogging potential.
- Avoid facial powders—use blotting papers instead to manage oiliness.
- Do not squeeze or pick at acne lesions.

Section 7. Global situation on bacterial resistance

Key points

- Global health authorities agree that antibiotic use should be minimized across all medical conditions to preserve effectiveness and reduce antimicrobial resistance.
- Antibiotics are commonly used in acne treatment, so it is essential for dermatologists and other clinicians to adopt antibiotic-sparing strategies.
- First-line therapy for acne is a combination of a topical retinoid and an antimicrobial agent; adding **benzoyl peroxide (BPO)** to the regimen helps reduce the risk of bacterial resistance.
- When systemic antibiotics are necessary, treatment duration should be limited.
- For women, hormonal therapy can be a valuable addition to topical treatments.

The problem of antimicrobial resistance and how it relates to acne

Antimicrobial resistance is rising worldwide due to widespread antibiotic use, both appropriate and inappropriate. Healthcare providers are increasingly encouraged to limit antibiotic prescriptions when alternative treatments are available—a principle known as **antibiotic stewardship**—to preserve antibiotics for situations where they are truly needed. Educational initiatives by local, national, and global health organizations have successfully reduced inappropriate antibiotic use in various medical fields. Similar efforts to improve antibiotic prescribing practices are expanding in the Middle East. In 2017, Al Tawfiq from Saudi Arabia reported that an educational intervention using both provider training and public awareness campaigns led to measurable reductions in unnecessary antibiotic prescriptions.